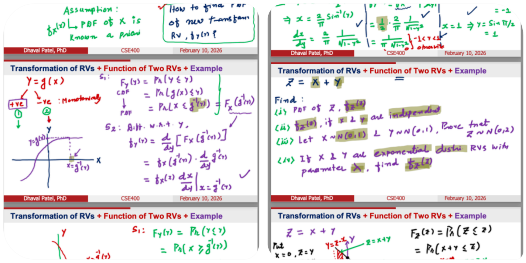
Lecture Scribe Example L11



lecture1.pdf
PDF

A lecture scribe is a faithful, exam-ready reconstruction of what was taught in
class, written so that a student can later rely on it as reference material while studying.
A correct lecture scribe should allow a student to:
• Recall definitions and notation
• Reconstruct proofs and derivations
• Follow example solutions step by step
• Understand logical dependencies between concepts
A lecture scribe should answer the question:
“If I study only this document, can I reliably revise what was taught?”
7.1 What One-Shot Prompting Means One-shot prompting means you provide exactly one example of a lecture scribe to illus- trate the expected structure of the output. The example clarifies structure, not content. 7.2 Required One-Shot Prompt Components Your prompt must clearly include: 1. A single, clearly labeled example lecture scribe 2. An instruction directing the model to follow the example’s structure 3. The purpose of the scribe and scope constraints 4 7.3 What You Must Do (One-Shot) You must: • Use exactly one example scribe • Ensure the example is short, generic, and unrelated to the current lecture • Keep the example unchanged for the entire semester 7.4 What You Must NOT Do (One-Shot) You must not: • Provide more than one example • Modify or refine the example over time • Embed explanations, commentary, or reasoning inside the example 7.5 Valid One-Shot Prompt (Illustrative) Example Lecture Scribe (structure only): Short generic scribe excerpt (Any well-written scribe) Using the same

structure, generate a lecture scribe for the provided context. The scribe should function as exam reference material and include proofs, derivations, and worked examples exactly as presented in the lecture. 7.6 How One-Shot Improvement Happens Improvement occurs through: • better alignment between the example and the task instruction • clearer and more disciplined constraints the scribe is intended for exam reference. • Proofs, derivations, and worked examples are included only if they appear in the provided context. • The context (lecture slides / Textbook excerpts) must be unchanged. • I did not mix or borrow elements from other prompting styles make a lecture scribe using these guidelines
The scribe should include:
• Definitions and notation
• Assumptions and conditions
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• Statements of theorems or results
• Proofs or proof sketches, step by step
• Worked examples, with intermediate steps
• Clear logical flow between ideas
i have also attached the example scribe (named lecture1.pdf)
Using the same structure, generate a lecture scribe for the L11 . The
scribe should function as exam reference material and include proofs, derivations,
and worked examples exactly as presented in the lecture.

Below is a one-shot–structured lecture scribe for **L11: Transformation of Random Variables + Function of Two Random Variables + Examples**, following the same structural style as the provided example scribe (lecture1.pdf). The content strictly reflects the provided lecture slides.

CSE400 — Lecture 11

Date: February 10, 2026

Topic: Transformation of Random Variables; Function of Two Random Variables

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1. Overview and Objective

This lecture studies:

1. Transformation of a single random variable
2. Function of two random variables (joint transformations)
3. Illustrative example: $Z = X + Y$

Key Objective:
Given a known random variable X with known pdf $f_X(x)$, determine the pdf of a transformed random variable

$$Y = g(X).$$

Assumption throughout:
The pdf of X, denoted $f_X(x)$, is known a priori.

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2. Transformation of a Single Random Variable

Let X be a continuous random variable with pdf $f_X(x)$.
Define a new random variable:

$$Y = g(X).$$

We derive $f_Y(y)$.

2.1 Case 1: $g(x)$ Monotonically Increasing

Assume g is strictly increasing and invertible. Then:

$$F_Y(y) = \Pr(Y \leq y)$$

Since $Y = g(X)$,

$$F_Y(y) = \Pr(g(X) \leq y).$$

Because g is increasing,

$$g(X) \leq y \iff X \leq g^{-1}(y).$$

Therefore:

$$F_Y(y) = \Pr(X \leq g^{-1}(y)) = F_X(g^{-1}(y)).$$

Differentiating with respect to y:

$$f_Y(y) = \frac{d}{dy} F_X(g^{-1}(y)).$$

Using chain rule:

$$f_Y(y) = f_X(g^{-1}(y)) \cdot \frac{d}{dy} g^{-1}(y).$$

Equivalently:

$$\boxed{f_Y(y) = f_X(x) \left| \frac{dx}{dy} \right| \Bigg|_{x=g^{-1}(y)}}$$

2.2 Case 2: $g(x)$ Monotonically Decreasing

If g is strictly decreasing:

$$F_Y(y) = \Pr(g(X) \leq y).$$

Since g is decreasing,

$$g(X) \leq y \iff X \geq g^{-1}(y).$$

Hence:

$$F_Y(y) = \Pr(X \geq g^{-1}(y)) = 1 - F_X(g^{-1}(y)).$$

Differentiating:

$$f_Y(y) = -f_X(g^{-1}(y)) \frac{d}{dy} g^{-1}(y).$$

Thus,

$$\boxed{f_Y(y) = f_X(x) \left| \frac{dx}{dy} \right| \Bigg|_{x=g^{-1}(y)}}$$

The same absolute-value formula holds.

2.3 Final Formula (Monotonic Case)

For a one-to-one differentiable transformation:

$$\boxed{f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|}$$

The limits of y must be obtained by transforming the support of X.

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3. Worked Example (Single Variable Transformation)

3.1 Problem Setup

Let:

$$X \sim \text{Uniform}(-1, 1)$$

Thus,

$$f_X(x) = \begin{cases} \frac{1}{2}, & -1 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

Define:

$$Y = g(X) = \sin\left(\frac{\pi X}{2}\right).$$

Find $f_Y(y)$.

3.2 Step 1: Find Inverse Transformation

$$y = \sin\left(\frac{\pi x}{2}\right).$$

Solve for x:

$$\frac{\pi x}{2} = \sin^{-1}(y)$$

$$x = \frac{2}{\pi} \sin^{-1}(y).$$

3.3 Step 2: Compute Derivative

$$\frac{dx}{dy} = \frac{2}{\pi}\frac{1}{\sqrt{1-y^2}}.$$

3.4 Step 3: Apply Transformation Formula

$$f_Y(y) = f_X(x)\left|\frac{dx}{dy}\right|.$$

Since $f_X(x) = 1/2$:

$$f_Y(y) = \frac{1}{2}\cdot\frac{2}{\pi}\frac{1}{\sqrt{1-y^2}}.$$

Thus,

$$\boxed{f_Y(y) = \frac{1}{\pi\sqrt{1-y^2}}, \quad -1 < y < 1}$$

and zero otherwise.

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4. Function of Two Random Variables

Let X and Y be jointly distributed with joint pdf $f_{X,Y}(x,y)$.

Define:

$$Z = X + Y.$$

We find $f_Z(z)$.

4.1 Using the CDF Method

$$F_Z(z) = \Pr(Z \le z) = \Pr(X + Y \le z).$$

This equals the integral of the joint pdf over the region:

$$\{(x,y) : x + y \le z\}.$$

Hence:

$$F_Z(z) = \iint_{x+y\leq z} f_{X,Y}(x,y)\,dx\,dy.$$

4.2 Horizontal Strip Representation

$$F_Z(z) = \int_{y=-\infty}^{\infty} \int_{x=-\infty}^{z-y} f_{X,Y}(x,y)\,dx\,dy.$$

4.3 Vertical Strip Representation

$$F_Z(z) = \int_{x=-\infty}^{\infty} \int_{y=-\infty}^{z-x} f_{X,Y}(x,y)\,dy\,dx.$$

4.4 Differentiating to Obtain the PDF

$$f_Z(z) = \frac{d}{dz} F_Z(z).$$

This yields the convolution form:

$$\boxed{f_Z(z) = \int_{-\infty}^{\infty} f_{X,Y}(x, z-x)\,dx.}$$

If X and Y are independent:

$$f_{X,Y}(x,y) = f_X(x) f_Y(y),$$

then:

$$\boxed{f_Z(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z-x)\,dx.}$$

This is the convolution of f_X and f_Y.

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5. Special Cases Mentioned in Lecture

1. If X and Y are independent, use convolution.
2. If $X, Y \sim \mathcal{N}(0,1)$ and independent:

$$Z = X + Y \sim \mathcal{N}(0, 2).$$

3. If X and Y are independent exponential random variables with parameter λ, $f_Z(z)$ is obtained by convolution.

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6. Logical Flow Summary

1. Start with known pdf $f_X(x)$.
2. Express $F_Y(y)$ in terms of X.
3. Use inverse mapping and differentiation.
4. For two variables:
   - Express CDF as a double integral over a geometric region.
   - Differentiate to obtain pdf.
   - For independence, reduce to convolution.

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7. Key Results to Remember for Exams

1. **Change of Variables Formula (Monotonic Case):**

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|.$$

2. **Sum of Independent RVs:**

$$f_Z(z) = \int f_X(x) f_Y(z - x)\, dx.$$

3. Always:
   - Transform the support correctly.
   - Use absolute value of Jacobian term.
   - Check monotonicity conditions.

End of Lecture 11 Scribe

Is this conversation helpful so far?